



Introduction to Python

By the end of this lesson, you should be able to:

- Identify and use common Python datatypes, including strings, integers, floats, and Booleans.
- Create, access, and change variables to store information in a program.
- Use `print()` to display values in the console.
- Use clear and descriptive names when creating variables.

Displaying Output with `print()`

The `print()` function displays information in the console.

Anything inside quotation marks is treated as text.

```
print("Hello World")
```

OUTPUT

```
Hello World
```

Variables

A **variable** is a named location that stores a value.

Variables allow programs to remember information and use it later.

Think of a variable as a labeled container: the variable name is the label, and the value is what is stored inside.

```
score = 100  
name = "Alex"  
  
print(score)  
print(name)
```

```
OUTPUT  
100  
  
Alex
```

Variables vs. Literals

A **literal** is a value written directly in your code.

A **variable** stores a value under a name.

Both display **15**, but the variable can be reused or changed later.

```
age = 15
```

```
print(age)    # Variable
```

```
print(15)     # Literal
```

OUTPUT

15

15

Datatypes

A **datatype** describes what kind of information a value contains.

The datatype determines how Python can use the value.

To the right are the most common primitive datatypes.

```
name = "Alex"      # str
age = 15            # int
height = 1.72       # float
is_student = True   # bool
```

Strings (**str**)

A **string** stores text as a sequence of characters.

Strings must be surrounded by quotation marks.

Numbers surrounded by quotation marks are also treated as strings rather than numbers.

```
name = "Alex"  
message = "Hello World"  
number = "42"
```

Numbers

Python commonly uses two datatypes for numbers.

An `int` stores a whole number, while a `float` stores a number with a decimal value.

Python determines which datatype to use based on the value being stored.

```
students = 24    # int  
temp = 21.5      # float
```

Integers (**int**)

An **integer** is a whole number without a decimal point.

Integers can be positive, negative, or zero.

They are useful for values that are counted using whole numbers.

```
age = 16  
score = 100  
temperature = -5
```


Floats (float)

A float is a number that contains a decimal point.

Floats are useful for measurements and other values that may contain a fractional or decimal amount.

```
price = 9.99  
height = 1.72  
temperature = -3.5
```

Booleans (bool)

A Boolean stores one of two possible values: True or False.

Booleans are commonly used to represent whether a condition is currently true or false.

In Python, True and False must begin with capital letters.

```
is_student = True  
game_over = False
```

Initializing Variables

Initializing a variable means creating it and assigning its first value.

Python uses the equals sign (=) to assign a value to a variable.

The variable name goes on the left of =, and the value being stored goes on the right.

```
score = 100  
player_name = "Alex"
```

Naming Variables

Variable names should clearly describe the information they store.

Python variables should normally use `snake_case`, with lowercase words separated by underscores.

Use descriptive words instead of unclear abbreviations or single-letter names.

```
player_score = 100  
student_name = "Alex"  
number_of_lives = 3
```

Accessing Variables

Accessing a variable means using the value currently stored inside it.

Use the variable's name whenever you want to access its value.

```
score = 100  
print(score)
```

OUTPUT

100

Changing Variables

A variable can be changed by assigning a new value to it.

The new value replaces the value that was previously stored.

```
score = 100  
print(score)
```

```
score = 90  
print(score)
```

OUTPUT

100

90

Putting It Together

Variables allow programs to store information, access that information later, and change it as the program runs.

Each variable has a name, stores a value, and has a datatype based on the value being stored.

```
player_name = "Alex"  
score = 100  
game_over = False  
  
print(player_name)  
print(score)  
  
score = 150  
print(score)
```



Congratulations!

You have completed the introduction to Python!

You should now be able to create variables using appropriate datatypes, access and change their values, and display information using `print()`.